

SOCIAL MEDIA SENTIMENT AS A PREDICTOR OF STOCK RETURNS

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Tavishka Arora & Owen Scott
Students, The University of Texas at Austin

GUIDING QUESTION



Does sentiment on social media about a stock predict its short-term price movements, or is it simply a reaction to them?

Can the collective sentiment expressed in social media discussions serve as a reliable predictor of short-term stock price movements?



Significance

Traditional financial models often overlook how retail investor behavior, amplified by social media, can influence markets. The 2021 “meme stock” episode showed that coordinated activity on platforms like Reddit’s WallStreetBets can meaningfully move prices and disrupt conventional market expectations. This marks a shift in market dynamics where social sentiment occasionally competes with, or even outweighs, traditional valuation signals.

1. **Market Democratization**
2. **Information Speed**
3. **Risk Management**
4. **Alternative Data**
5. **Regulatory Considerations**

DATA SOURCES

Original Sources

- **Reddit WallStreetBets Posts (53K raw posts)**
 - Post titles, content, scores, comment counts, timestamps
 - Time period: January 2021 (peak meme stock era)
 - Allowed us to extract Rich behavioral and sentiment signals from retail investors
- **Historical Stock Price Data (Yahoo Finance API)**
 - Daily adjusted closing prices for all mentioned tickers
 - ± 7 day price windows around each post date
- **Corporate Event Data (Yahoo Finance)**
 - Earnings & ex-dividend dates
 - Dividend distributions and stock splits

Data Integration Process

These heterogeneous data sources were systematically merged through a multi-step pipeline:

1. Sanitize post title/body and merge them into one 'text'

1. Remove: Usernames, URLs, mentions, etc.

2. Label all posts using tickers

1. pulled from `pytickersymbols` + Custom Additions

3. Pull stock pricing

1. Data ranging from T-7 → T+7

4. Create EEID List

1. Pull earnings, ex-dividend, dividend payout, and stock split dates from Yahoo Finance
2. These events unpredictably influence the market and data from these dates should be ignored

5. Perform sentiment analysis on each post:

1. Used our WSB-customized sentiment analysis tool as well as FinBERT, the existing SOTA transformer

6. Average each sentiment strategy per-ticker per-day for all posts for more robust sentiment metrics and reduced noise.

The final composite dataset joins social sentiment signals with corresponding price movements and contextual market information, creating a comprehensive feature space that captures both behavioral and fundamental market dynamics. prediction modeling.

Data Preprocessing & Composite Dataset Creation

1. **Temporal Alignment Challenge:** Posts occur at any time; stock prices are daily snapshots
2. **Ticker Extraction & Validation:** Raw posts mention multiple tickers in unstructured text
3. **EEID-Aware Feature Engineering:** Stock prices around earnings/events are inherently noisy
4. **Price Window Construction:** Missing trading days, weekends, holidays
5. **Data Quality & Completeness:** Final Dataset: 39,124 observations & 30 columns after filtering

CURRENT STATE-OF-THE- ART & OUR APPROACH

Existing Literature & SOTA Methods

Traditional Financial Sentiment Analysis

1. Loughran–McDonald Dictionary (2011) - Standard lexicon for financial filings (10-Ks, earnings statements).
 2. VADER (Hutto & Gilbert, 2014) - Rule-based social media sentiment; not finance-specific.
 3. TextBlob / AFINN - General sentiment tools used in early financial text studies.
 4. FinBERT (Araci, 2019) - Transformer model trained on financial reports; stronger contextual understanding than lexicons.
- Limitation:** Tools designed for formal financial language, not retail investor slang or meme-driven communities.

Social Media Finance Research (Twitter Focused)

1. Bollen et al. (2011) - "Twitter mood predicts the stock market" using OpinionFinder
 1. Basically Proven Void
 2. Zhang et al. (2011) - Daily S&P 500 prediction using Twitter sentiment
 3. Ranco et al. (2015) - Twitter sentiment correlation with Dow Jones movements
- Limitation:** general sentiment dictionaries not adapted to retail investor slang

Reddit Analysis

1. Stoeckl & Huber (2020) - Reddit sentiment applied to crypto markets (not equity-focused)
 2. Archak et al. (2011) - Product-review sentiment (non-financial)
 3. Mittal & Goel (2012) - News-driven prediction (pre-Reddit era)
- Limitation:** No systematic analysis of Reddit's finance-focused communities, particularly regarding how their unique language and discussion dynamics align with or diverge from traditional sentiment signals.

Existing Literature & SOTA Methods

Prediction Horizons

1. Makridakis & Hibon (2000) - M-competitions focus on weekly/monthly forecasts
2. Timmermann (2006) - "Forecast combinations" for monthly returns
3. Patel et al. (2015) - Daily price prediction using technical indicators

Limitation: Systematic comparison across 1-7 day horizons using social sentiment signals

EEID Considerations

1. Patell & Wolfson (1984) - Event studies around earnings announcements
2. MacKinlay (1997) - Event study methodology framework

Limitation: Integration with social media sentiment analysis during volatile periods

Modern Deep Learning Approaches

1. FinBERT - Domain-specific transformer pretrained on financial corpora
2. Temporal Convolutional Networks (TCN) - Strong for time-series modeling, but underused in financial sentiment + prediction pipelines.

Limitation: Limited research combining FinBERT sentiment + TCN price forecasting, especially using Reddit data.

Our Approach: Hybrid Sentiment Analysis for Multi-Horizon Stock Prediction

1. Domain-Specific Sentiment Engineering

1. Custom WSB Lexicon
2. Dual-Method Comparison

Contribution: First systematic incorporation of WSB-style terminology into a hybrid sentiment framework for financial prediction.

2. Expected External Influential Dates (EEID) Considered in Framework

1. List of: Earning Dates, Ex-Dividend Dates, Stock Split Dates, Dividend Payout Dates
2. By excluding dates associated with major market-moving events, our model produced noticeably stronger and more stable performance.

Contribution: First application of EEID aware sentiment modeling for social media data

3. Systematic Multi-Horizon Analysis

1. Prediction Targets: Day 0 + [1, 2, 3, 4, 5, 6, 7]; 1, 3, 7 Reported
2. Input Features: [Sentiment, Price History, Both]
3. 5 EEID filtering configurations
4. Model Architectures: [Linear, Ridge, Random Forest, XGBoost, LSTM, Hybrid CNN/LSTM, LightGMB, TCN]
5. Performance Metrics: [RMSE, Sign-Accuracy]

No existing studies combine WSB-specific sentiment analysis with systematic multi-horizon prediction while accounting for EEID - representing a significant methodological and empirical contribution to the field. This work represents a novel application of existing sentiment analysis and machine learning techniques to an unexplored problem domain (Reddit WSB during meme stock era) with custom innovations in domain-specific feature engineering and EEID-aware modeling.

MODELS COMPARISON & EVALUATION



PRICE-ONLY MODEL



Price-Only Model

Objective: Investigate inverse market dynamics by predicting future WSB community sentiment reactions from current price movements, establishing causal relationships between market events and social media response.

Feature Engineering (8 Input Features):

1. Price Momentum
2. EEID Measures
3. Temporal Features

Modeling Approaches Tested:

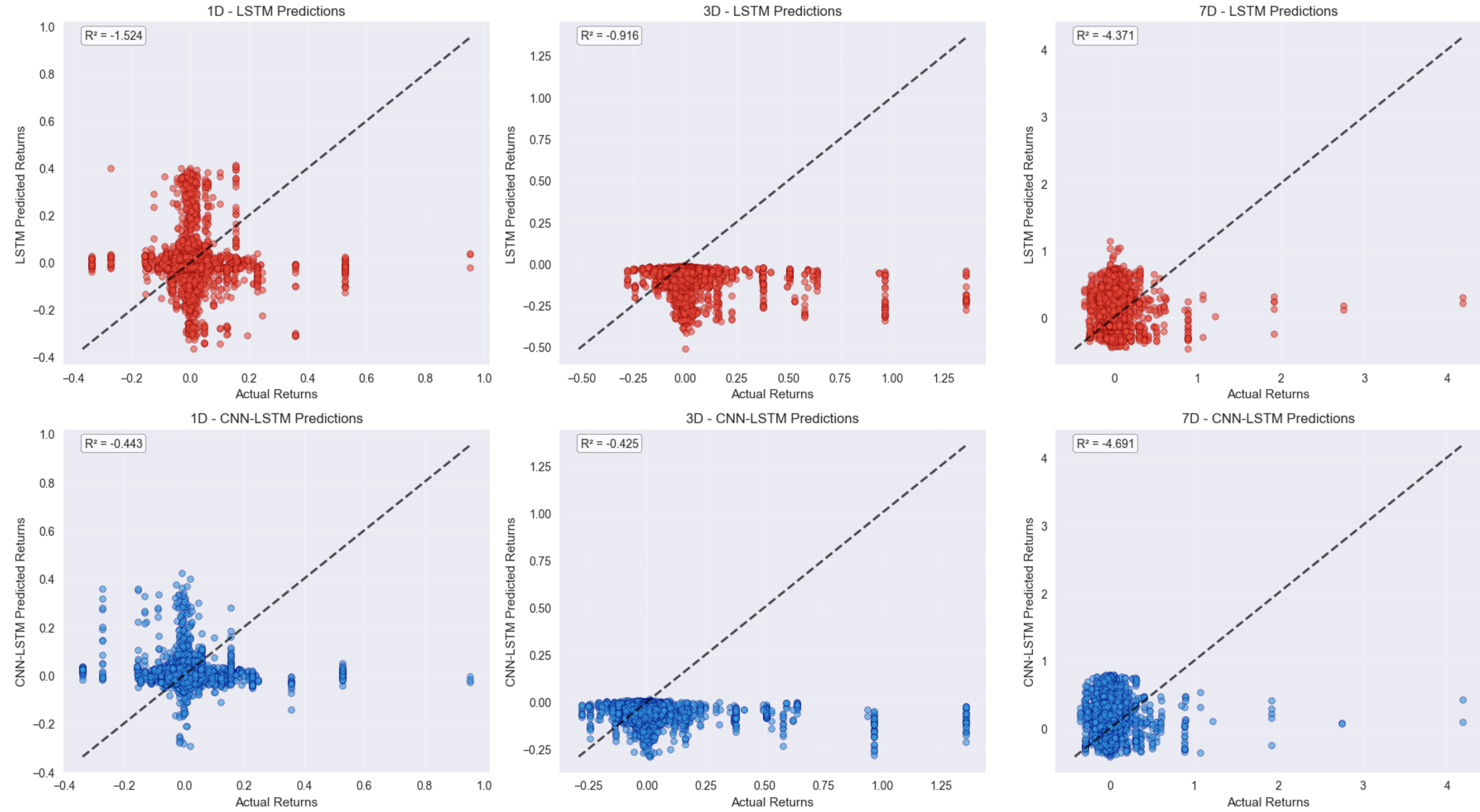
1. Traditional Machine Learning:
 1. Ridge Regression
 2. Random Forest
 3. Linear Regression
2. Deep Learning Attempts:
 1. LSTM Networks
 2. CNN-LSTM Hybrid

Challenge: Limited sequential structure in price-only data led to poor performance

Model Performance Comparison



Price-Only Model



Price-Only Model

Critical Findings

1. Prediction decay

- 1. Performance deteriorates significantly with longer horizons

2. Traditional ML dominance

- 1. Deep learning failed due to insufficient sequential patterns in price data

3. Feature importance

- 1. Momentum indicators and EEID measures most predictive, although overall model's predictive power is not significant.

4. Baseline establishment

- 1. Close to random predictive power from technical features alone

What This Tells Us

- 1. Price patterns contain limited predictive information - consistent with efficient market theory
- 2. Short-term technical signals exist but are weak and noisy
- 3. Deep learning requires richer feature sets than pure price data provides

This establishes that while traditional technical analysis has some merit on an experimental level, the predictive power is very limited, creating a compelling case for incorporating alternative data sources like social media sentiment.

PRICE PREDICTS FUTURE SENTIMENT MODEL



Price Predicts Future Sentiment Model

Objective: Investigate inverse market dynamics by predicting future WSB community sentiment reactions from current price movements, exploring relationships between market events and social media response.

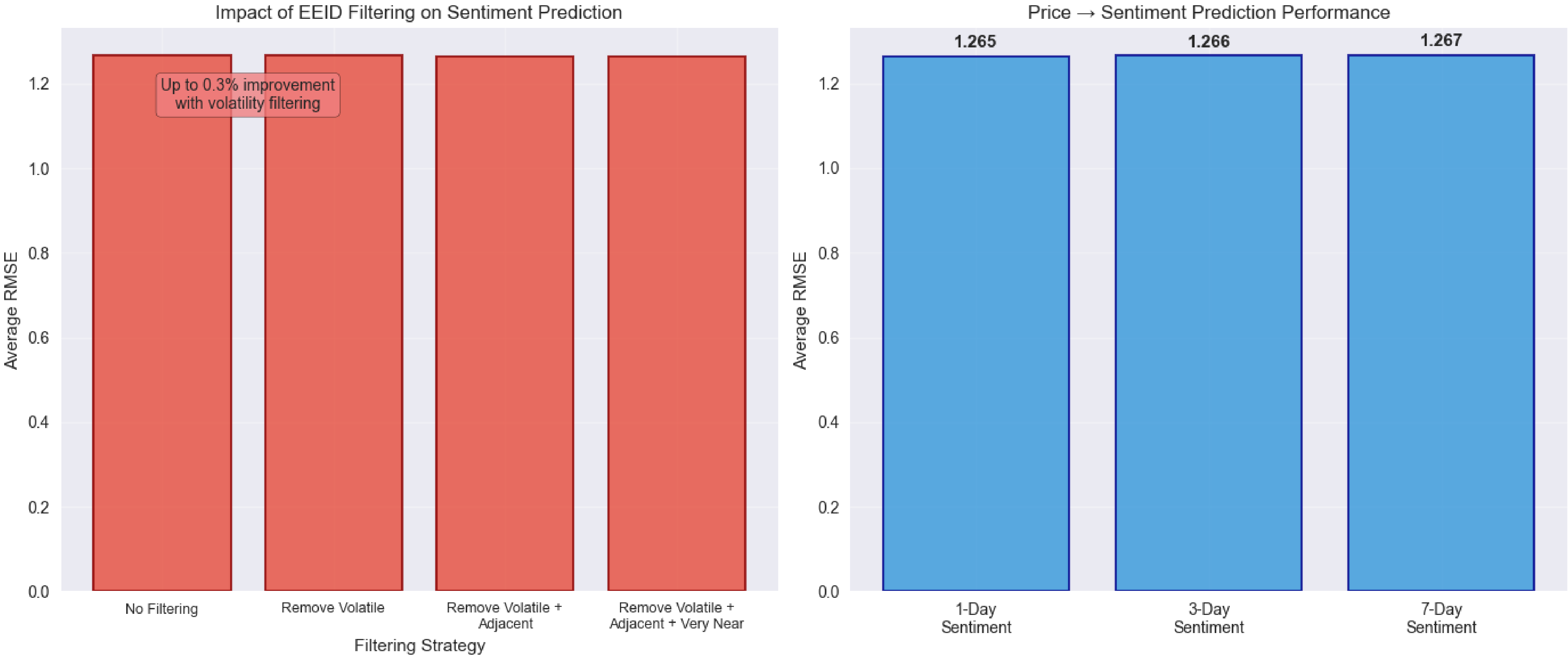
Feature Engineering (4 Core Features):

1. Current Return: Daily price movement (t+0 vs t-1)
2. 3-Day Momentum: Short-term price trajectory (t+0 vs t-3)
3. 7-Day Momentum: Medium-term price trends (t+0 vs t-7)
4. Price EEID: How much prices fluctuated over past 7 days

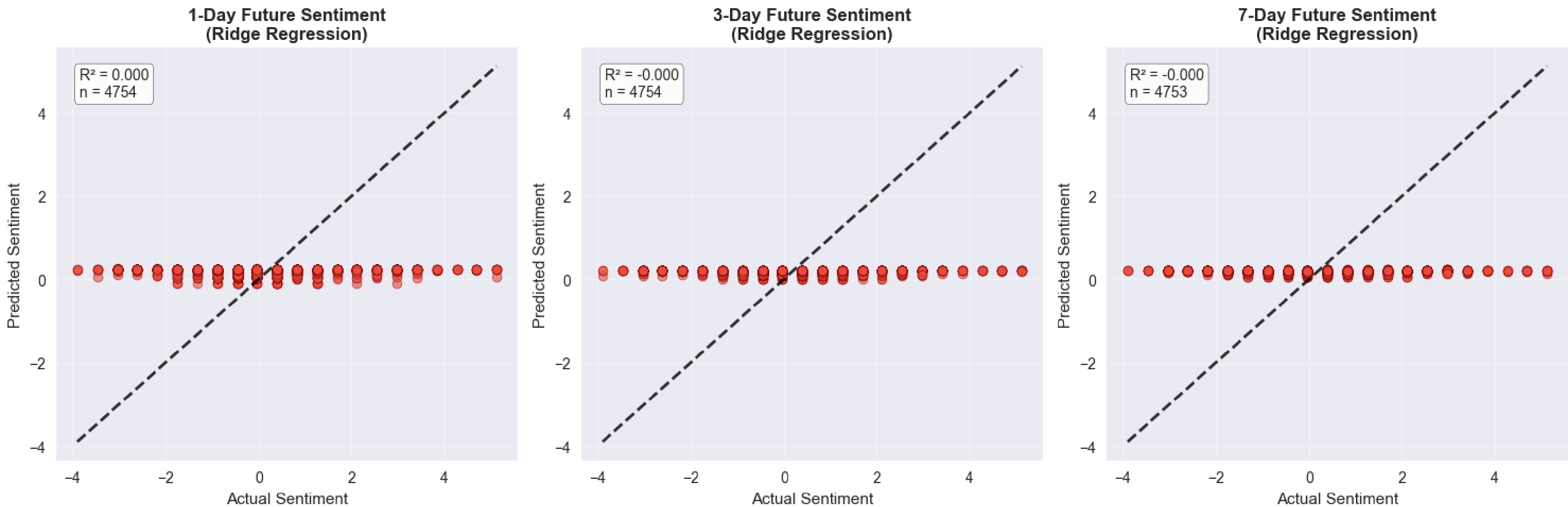
Best Model Performance

1. Algorithm: Ridge Regression with L2 regularization
2. Target: 3-day future Custom WSB sentiment (sentiment_3d)
3. RMSE: 1.213806
4. Correlation: 0.007 (p-value: 0.6720)
5. EEID filtering: 3 levels applied (removed volatile, adjacent volatile, and very near volatile dates)

Price Predicts Future Sentiment Model



Price Predicts Future Sentiment Model



Price Predicts Future Sentiment Model

Critical Findings:

1. Weak predictive signal from price movements to future sentiment: $R^2 \sim 0 \rightarrow$ These predictions are essentially no better than random guessing
2. EEID filtering provides minimal performance gains
3. Ridge regression slightly outperforms XGBoost
4. Relationships exist but with high uncertainty
5. Short-term predictions are marginally more reliable, yet not significant due to weak correlation (.007)

What This Tells Us:

1. Virtually no evidence of price $\rightarrow$ sentiment relationships, but effects are small
2. WSB community shows detectable but weak behavioral responses to price movements
3. Inverse causality has limited practical predictive value
4. Results suggest complex, multi-factor sentiment formation beyond just price

**SENTIMENT-ONLY MODEL:
SENTIMENT PREDICT
FUTURE PRICE**



Sentiment-Only Model: Sentiment Predict Future Price

Objective: Leverage Reddit WallStreetBets sentiment to predict stock returns, combining custom domain-specific lexicon with transformer-based sentiment analysis.

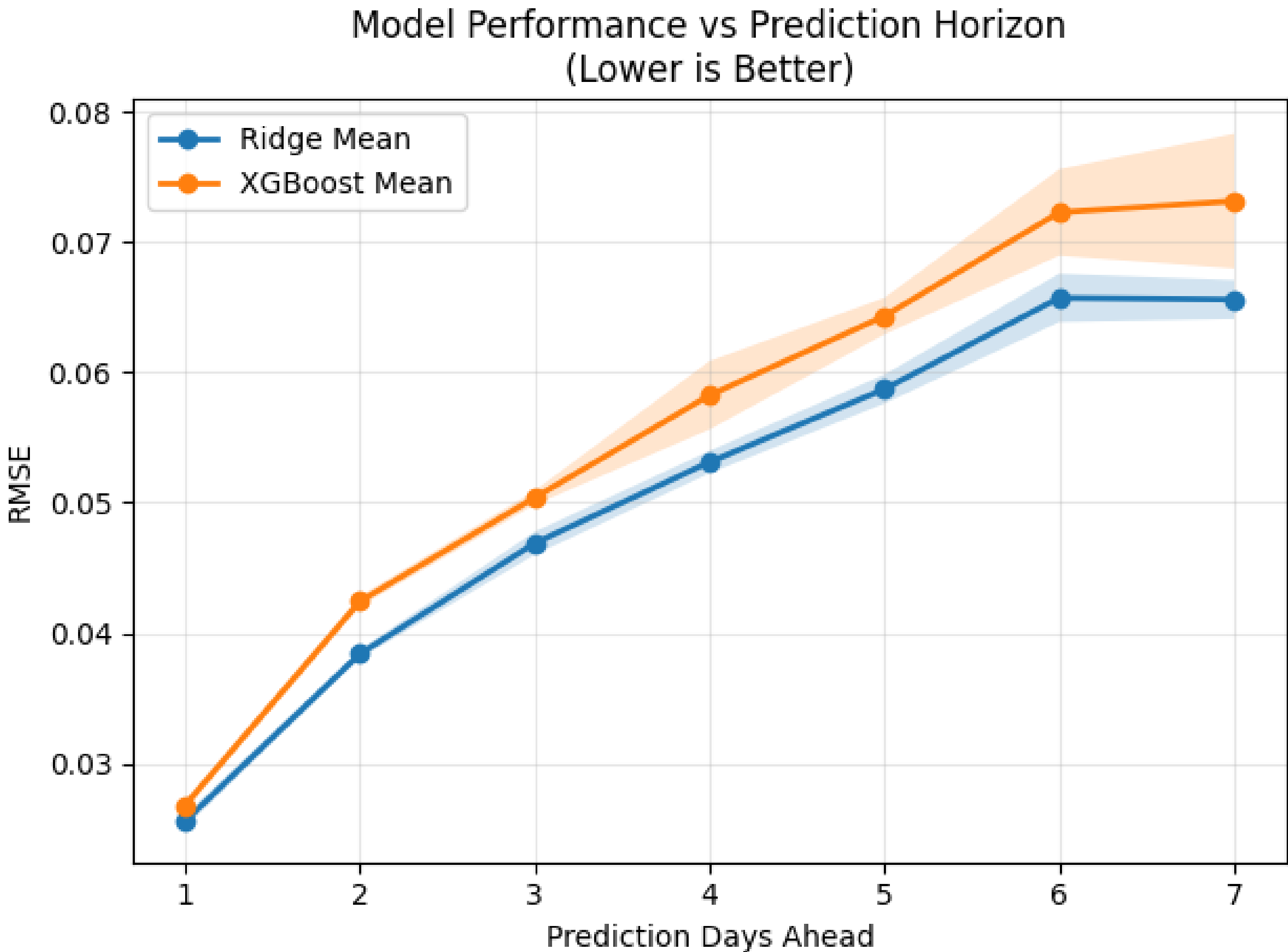
Feature Engineering (2 Core Features):

1. Custom WSB Lexicon: 50+ community-specific terms ("tendies," "diamond hands," "to the moon")
2. FinBERT Sentiment: Transformer-based financial sentiment scoring
3. EEID-Aware Filtering: 5-level system removing observations near earnings/ex-dividend dates, stock splits, dividend payouts
4. Multi-Horizon Targets: 1, 3, and 7-day return predictions
5. Grid Search Framework: 70 model configurations tested systematically

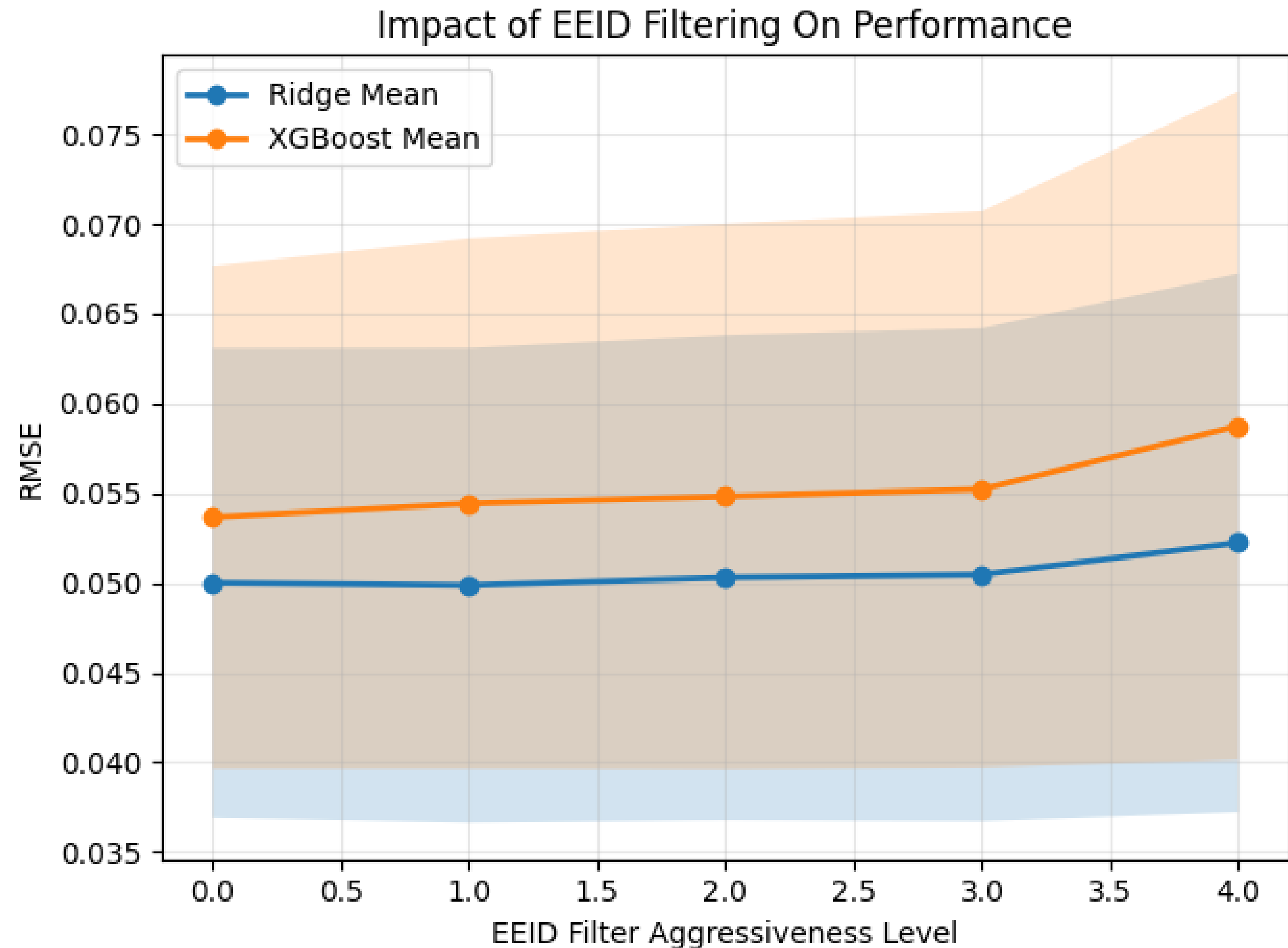
Modeling Approaches Tested:

1. Traditional Machine Learning:
 1. Ridge Regression: L2 regularization with dual sentiment inputs
 2. XGBoost: Gradient boosting with custom hyperparameters
 3. Systematic EEID filtering from no filtering to comprehensive event removal
2. Sentiment Used:
 1. Custom WSB + FinBERT Dataset

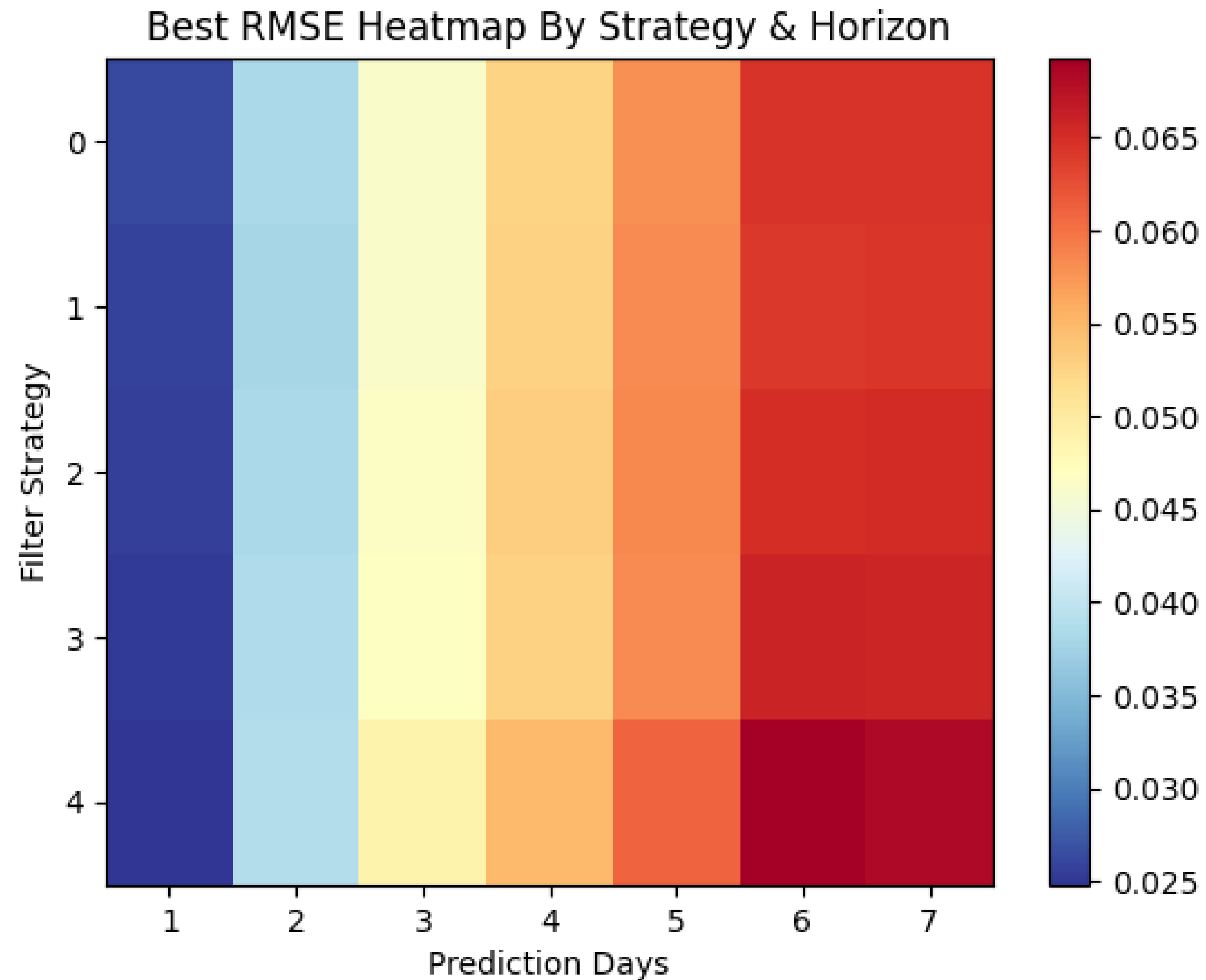
Sentiment-Only Model: Sentiment Predict Future Price



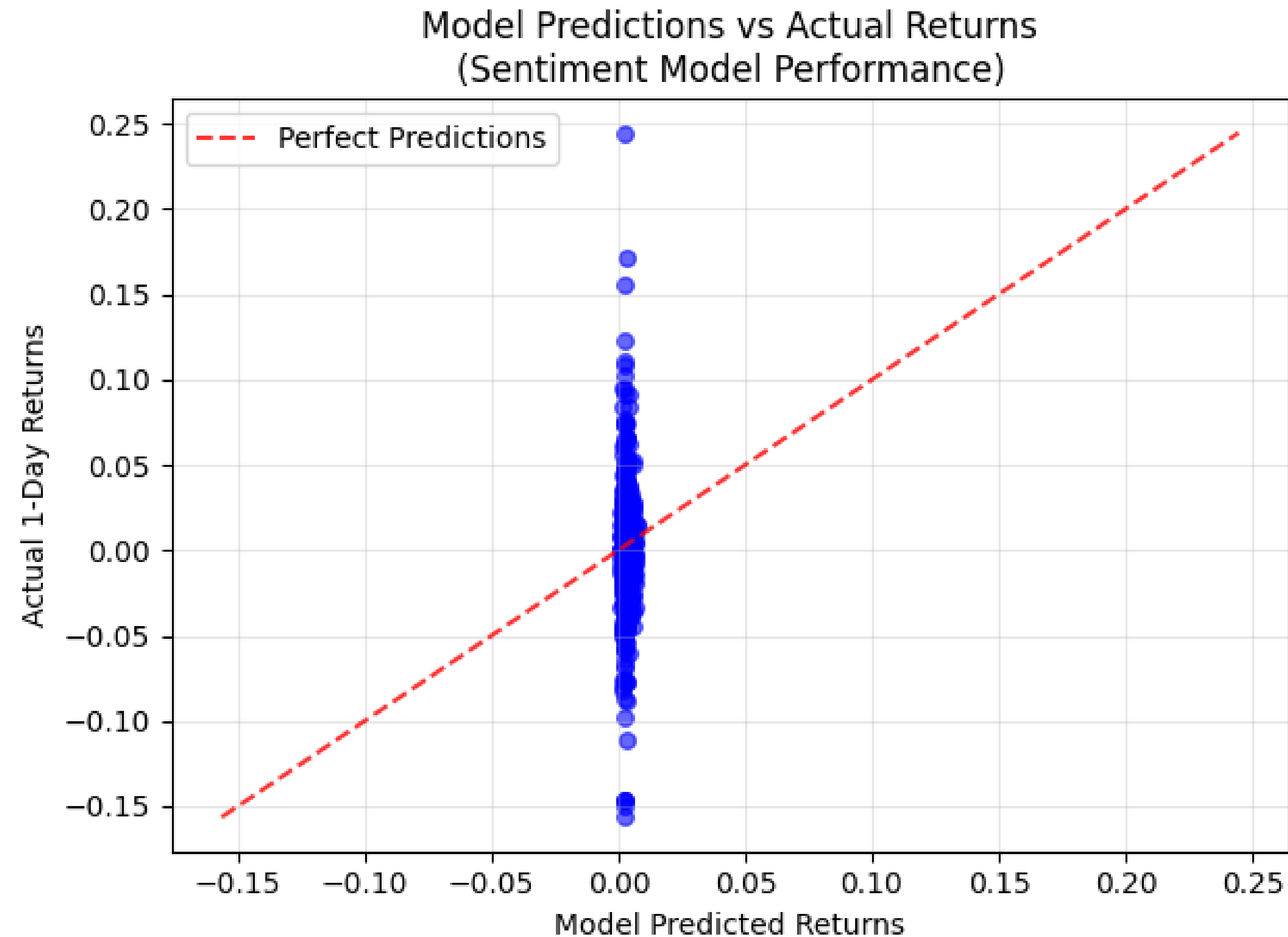
Sentiment-Only Model: Sentiment Predict Future Price



Sentiment-Only Model: Sentiment Predict Future Price



Sentiment-Only Model: Sentiment Predict Future Price



Sentiment-Only Model: Sentiment Predict Future Price

Critical Findings:

1. EEID filtering provides a very conservative performance improvement
2. Directional Accuracy is essentially random (47.1%)
3. Short-term predictions (1-day) most reliable across sentiment methods
4. Ridge regression outperforms XGBoost consistently
5. Sentiment signals are not enough to predict the market well

What This Tells Us:

1. WSB sentiment contains minimal exploitable signal (correlation ~ 0)
2. Retail sentiment is not predictive
3. Market efficiency largely dominates: easily accessible sentiment data provides little to no edge
4. Short-term noise dominates any sentiment signal (all prediction horizons struggle)

Our model does not perform better than random guessing for stock direction and shows no economically significant predictive power. This is the expected result in efficient markets and validates existing financial theory.

HISTORICAL PRICE + SENTIMENT MODEL



Historical price + Sentiment Model

Objective: Leverage Reddit WallStreetBets sentiment to predict stock returns, combining custom domain-specific lexicon with transformer-based sentiment analysis. The goal is to predict multi-horizon stock returns (1-7 days) using both historical price data and sentiment features.

Feature Engineering (10 Core Features):

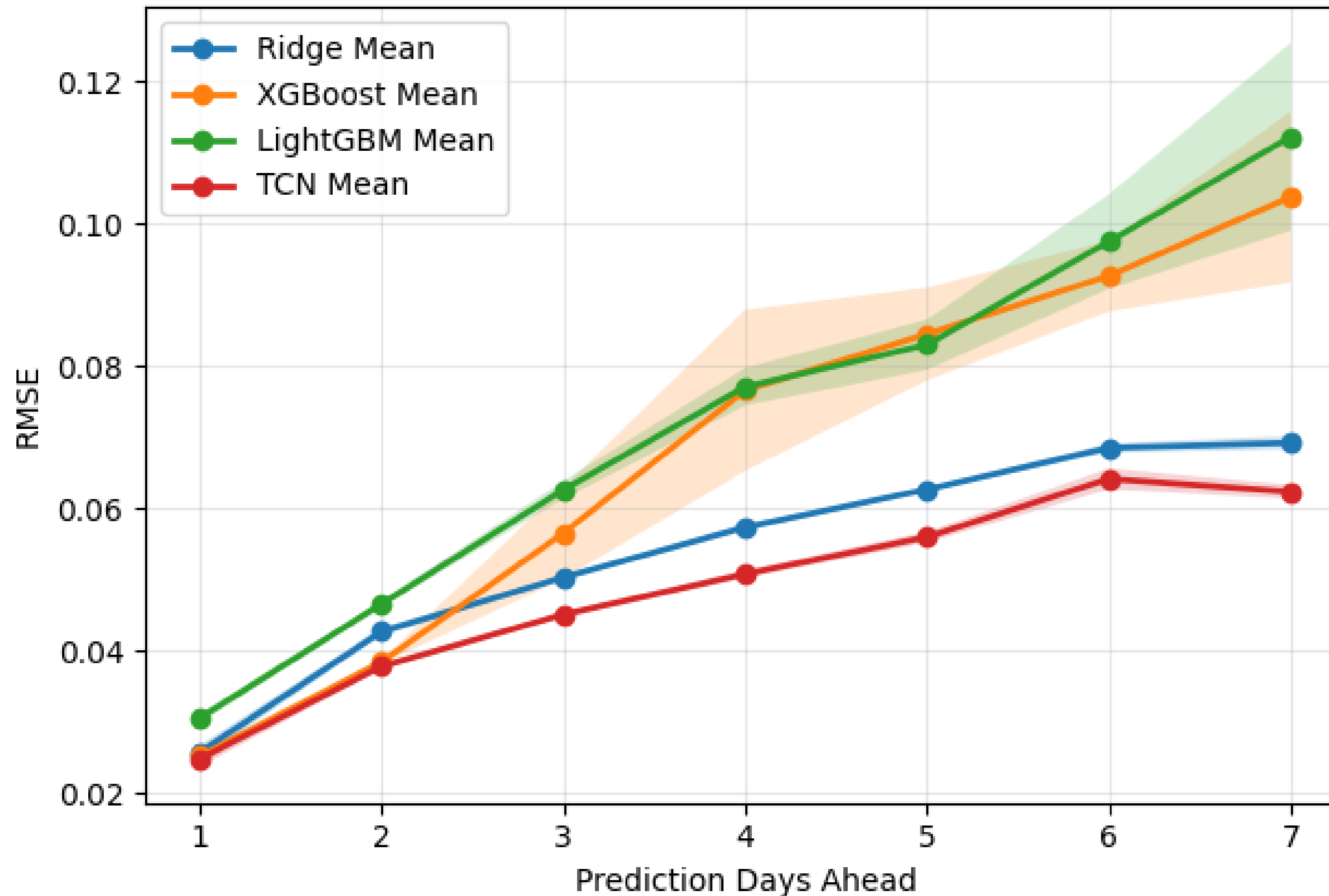
1. Price Features (8 features): Historical stock prices from 7 days before to current day (p-7 through p+0)
2. Sentiment Features (2 features):
 1. custom_sentiment: WSB-specific lexicon with 50+ community terms
 2. finbert_sentiment: Transformer-based FinBERT sentiment scoring
3. Target Variables: Return calculations for multiple horizons (r1 through r7 for 1-7 day predictions)
4. EEID-Aware Filtering: 5-level systematic filtering

Models Tested

1. **Ridge Regression:** L2 regularization with dual sentiment inputs and historical price inputs ($\alpha=1.0$)
2. **XGBoost:** Gradient boosting with hyperparameters:
 1. n_estimators=300, max_depth=3, learning_rate=0.05, subsample=0.8, colsample_bytree=0.8
3. **LightGBM:** Advanced gradient boosting with:
 1. num_leaves=31, learning_rate=0.03, regularization (L1=0.1, L2=0.1)
4. **Temporal Convolutional Network (TCN):** Custom deep learning architecture
 1. Channel format: [16, 32, 32] with weight normalization
 2. Kernel size=2, dropout=0.2, early stopping patience=7

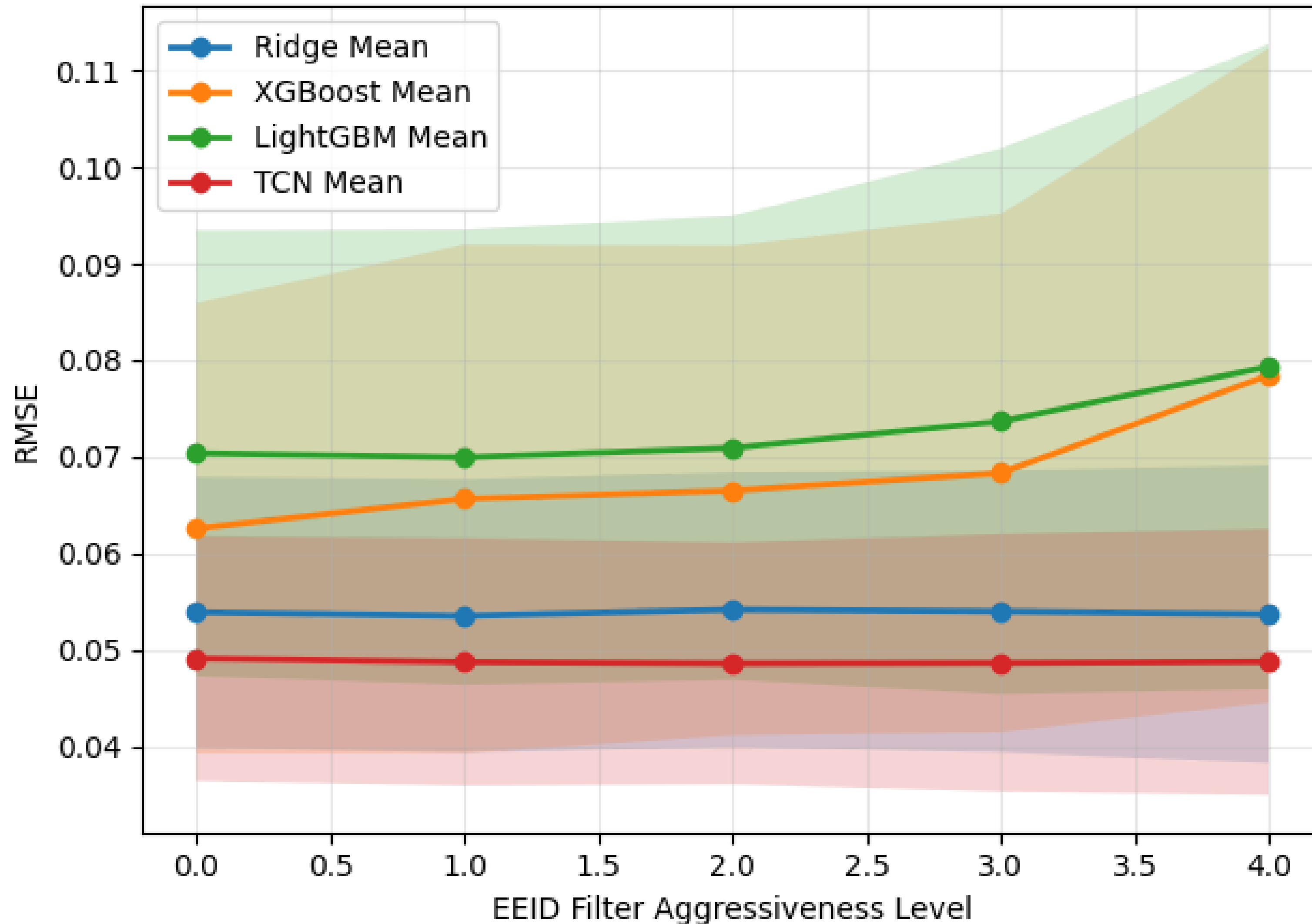
Historical price + Sentiment Model

Model Performance vs Prediction Horizon
(Lower is Better)



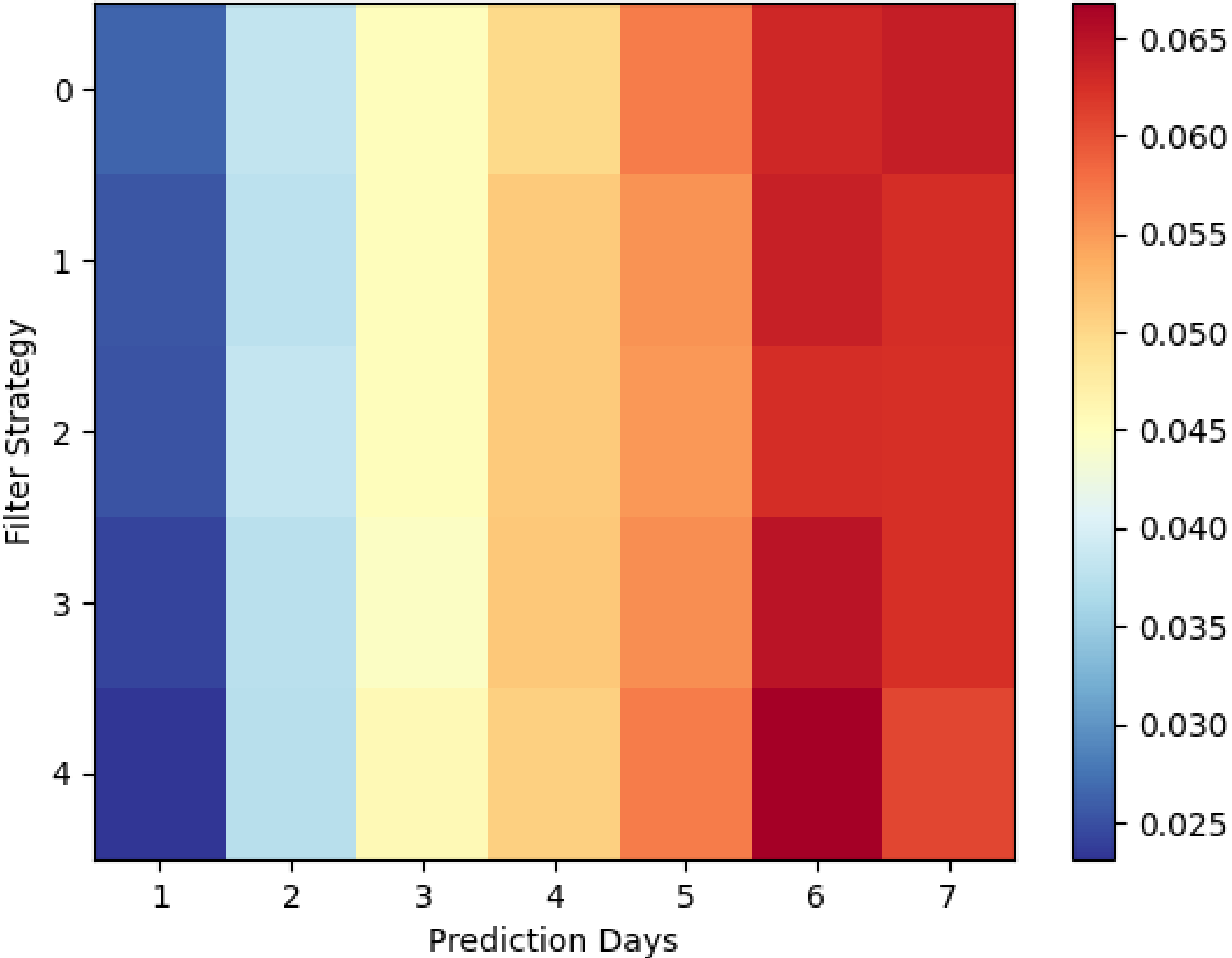
Historical price + Sentiment Model

Impact of EEID Filtering On Performance



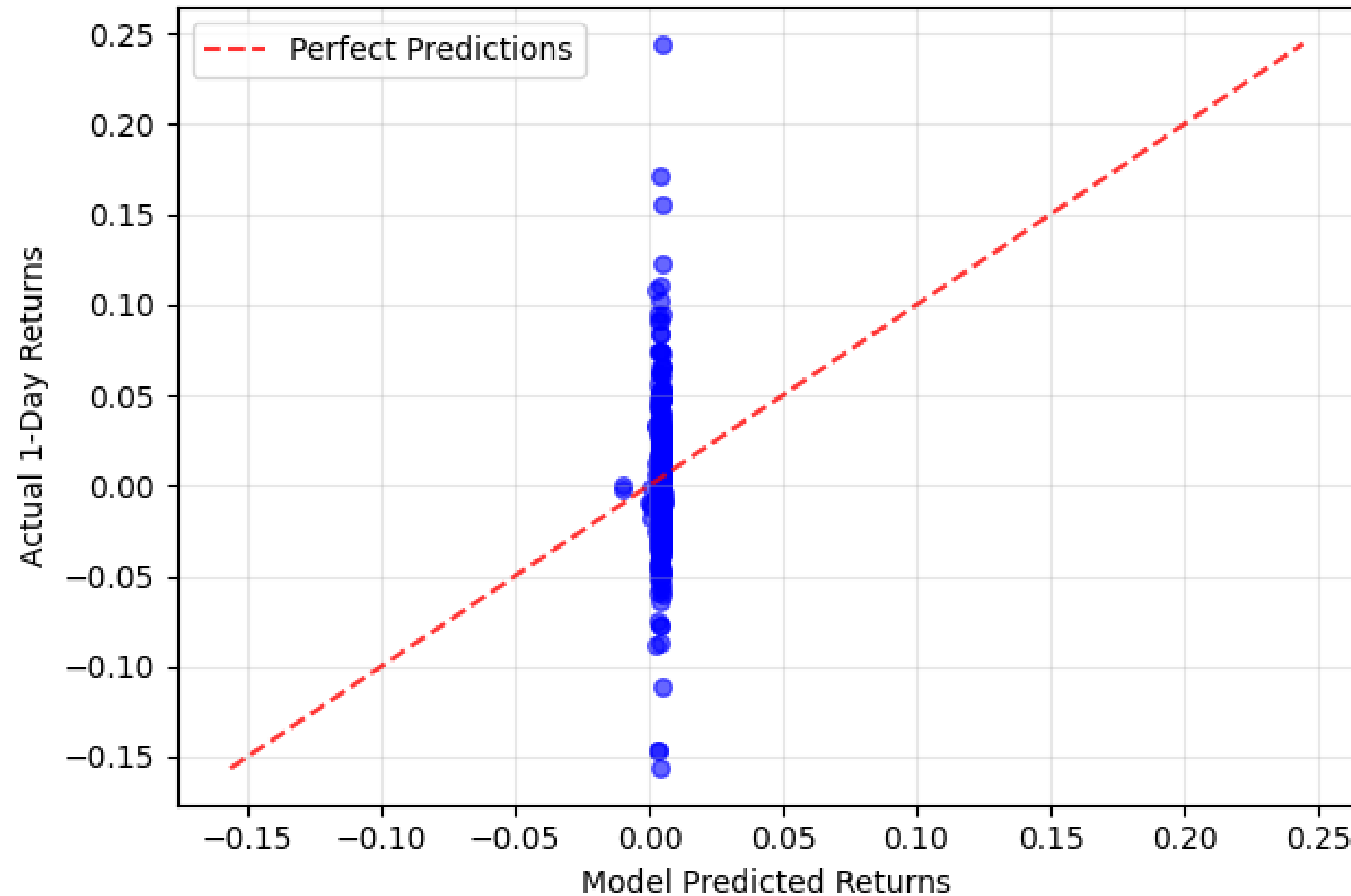
Historical price + Sentiment Model

Best RMSE Heatmap By Strategy & Horizon



Historical price + Sentiment Model

Model Predictions vs Actual Returns
(Sentiment + Prices Model Performance)



Historical Price + Sentiment Model: Sentiment Predict Future Price

Critical Findings:

1. EEID filtering provides a very conservative performance improvement
2. Directional Accuracy is essentially random (49.7%)
3. Short-term predictions (1-day) most reliable across sentiment methods
4. TCN outperforms alternative models significantly
5. Sentiment and short-history signals are not enough to predict the market well

What This Tells Us:

1. WSB sentiment contains minimal exploitable signal (correlation ~ 0)
2. Retail sentiment and short-term price history are not meaningfully predictive
3. Market efficiency largely dominates: easily accessible sentiment data provides little to no edge
4. Short-term noise dominates any sentiment signal (all prediction horizons struggle)

Our model does not perform better than random guessing for stock direction and shows no economically significant predictive power. This is the expected result in efficient markets and validates existing financial theory.

FINAL REMARKS



Conclusion

Our Motivation: Explore whether community sentiment from Reddit's WallStreetBets can meaningfully relate to short-term market behavior.

Our Approach: Built a structured pipeline that blends domain-specific slang, transformer-based sentiment, and event-aware filtering to examine sentiment and returns across multiple horizons.

What We Found: Sentiment in retail communities is highly noisy and influenced by many factors beyond price alone. While small patterns exist, they are not strong enough for reliable prediction.

What We Gained:

- A clearer understanding of the limits of sentiment-driven return prediction
- Practical ML insights: how noise affects model stability, how regularization behaves in low-signal settings, and why short-term horizons often yield fragile correlations.
- A scalable framework for deeper exploration of retail investor behavior.

Overall Future Direction: Shift focus toward richer sentiment signals, larger datasets, and more advanced temporal models to better capture the dynamics of retail-driven discussions.

Future Direction Using WSB: Explore predictive power of general buzz/ mention of a stock to predict price trends.

THANK YOU.



The University of Texas at Austin
Cockrell School of Engineering

Innovation starts **here**

